When Does Geometry Beat a Token-Entropy Baseline for Predicting Semantic Entropy? A Robustness Audit of Training-Free Hallucination Signals

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Abstract

Training-free geometric features of transformer hidden states—trajectory curvature, eigenscore, effective rank—have been proposed as cheap, probe-free signals of semantic uncertainty for hallucination detection. We audit whether they deliver rather than proposing a detector, and find that a trivial token-entropy baseline matches or beats every one of them, and that which features appear reliable is set by the evaluation split rather than by the geometry. Across Llama-3-8B-Instruct and Mistral-7B on TruthfulQA, BioASQ and SQuAD v2 (six model$\times$dataset cells), we score each feature as a direct predictor of meaning-clustered semantic entropy against the model's mean per-token entropy—one scalar from the same forward pass, computed independently of the answer sampling that defines the target, so the comparison is not circular. Token entropy wins on all five seeds in five of six cells, by a partial-correlation margin of $+0.116$ to $+0.191$: geometry fails to beat the baseline it was introduced to improve upon. Eigenscore and effective rank are largely sequence-length confounds—length explains $\sim$98–99% and 58–93% of their variance—and their apparent signal vanishes under nonlinear length control. Curvature, the one length-independent feature, is small (partial $r \approx 0.1$–0.18) and redundant with the baseline. At the pre-registered held-out endpoint ($n = 245$) the number of model$\times$dataset cells in which curvature's confidence interval excludes zero swings from zero to three across five splits, while the full sample ($n = 817$) is fixed at four. Because more than a third of items sit at the target's floor—all ten generations collapsing into a single meaning cluster—every correlation reported is an attenuated lower bound; this bounds absolute magnitudes but not the ordering between features, which face an identical target. We argue that any training-free geometric uncertainty feature should be required to clear a token-entropy baseline on equal footing and survive a reseeded held-out evaluation—two cheap checks that the features examined here do not pass.

1 Introduction

Large language models generate fluent text whose factual reliability they do not themselves flag, and detecting when a generation is likely a hallucination has become central to deploying them safely. A growing line of work pursues training-free uncertainty estimates: signals read directly from a model's forward pass, requiring neither a fitted probe nor multiple sampled generations. Among the most prominent candidates are geometric properties of the hidden-state trajectory—the curvature of the path a representation traces across layers (King et al., 2026; Damirchi et al., 2026), the eigenspectrum of its hidden states (eigenscore and effective rank; Chen et al., 2024; Wang et al., 2025), and related single-pass structural descriptors of representational stability (Yang et al., 2026). The appeal is concrete: these features are cheap, and several were introduced precisely because token-level confidence signals such as per-token entropy were judged too crude to capture semantic uncertainty (Kuhn et al., 2023; Farquhar et al., 2024). That motivation is rarely tested. Geometric uncertainty features are seldom evaluated against the very baseline they were meant to improve upon, and seldom stress-tested for whether a conclusion drawn from one held-out split survives a different one. Two assumptions thus go unexamined: that hidden-state geometry beats a trivial token-level signal, and that a single evaluation protocol reports a stable verdict about which features work. This paper audits both; we do not propose a detector. Across two models (Llama-3-8B-Instruct, Mistral-7B) and three datasets (TruthfulQA, BioASQ, SQuAD v2)—six model$\times$dataset cells—we measure how well each geometric feature predicts a hybrid semantic-entropy target (Kuhn et al., 2023; Farquhar et al., 2024), under partial-correlation controls for sequence length and its square. We are candid that representational-trajectory curvature was our own starting hypothesis: we set out to test whether hidden-state geometry could approximate semantic entropy cheaply. A free token-entropy baseline beat it. That outcome motivated the audit rather than serving as its centerpiece. Three findings follow. First, mean per-token entropy—a single number requiring no hidden states—matches or beats every geometric feature we test for predicting semantic entropy, winning on all five seeds in five of six cells with a partial-correlation gap of $+0.116$ to $+0.191$; geometry fails to beat the baseline it was introduced to improve upon. That a cheap token-level signal can rival elaborate semantic-uncertainty estimates is not itself new; what we add is the equal-footing, untrained, semantic-entropy-target comparison on which the geometric features—introduced precisely to improve on such signals—still do not win. Second, which features appear to carry genuine signal is unstable under resampling: at the held-out sample size the count of features whose confidence interval excludes zero ranges from zero to three across random splits, with no feature clean on every split, while the full-sample evaluation is fixed at four. The apparent survivor set is largely a resampling artifact rather than a reproducible fact. Third, the two most prominent spectral features are dominated by sequence length—eigenscore correlates with length at roughly 0.981–0.996 per cell ($\sim$98–99% of variance), and effective rank explains 58–93% of variance on length and its square—and their apparent uncertainty signal does not survive nonlinear length control. We state this correlationally: our data cannot separate length from an upstream factor such as question difficulty. The one geometric feature with a genuine length-independent signal—curvature—ties the audit together as an honest negative. Its effect is small (partial $r \approx 0.1$–0.18), CI-clean only on the pre-registered split of a single cell (SQuAD-Llama, $+0.172$ $[0.050, 0.297]$; controlling sequence length, length^2, and token entropy), redundant with the token-entropy baseline, and itself unstable across splits. Even the real geometric signal loses to the free one. We make four contributions:

- **(C-B)** A trivial token-entropy baseline matches or beats every training-free geometric uncertainty feature we test for predicting semantic entropy, across two models and three datasets (shown head-to-head for curvature, the one length-independent feature; eigenscore and effective rank are dispatched as length confounds, with no length-controlled signal to compare)—geometry fails to beat its own baseline.

- **(C-C)** This ordering is protocol-dependent and resampling-unstable: held-out conclusions about small geometric effects are not reproducible across random splits, and the survivor set is largely an artifact of evaluation design at these sample sizes.

- **(C-A)** Two prominent spectral features, eigenscore and effective rank, are largely explained by sequence length and do not survive nonlinear length control; we report this correlationally, noting that length and upstream question difficulty are not separable in our data.

- **(C-D)** The sole geometric feature with a genuine length-independent signal, curvature, is small, redundant with the baseline, and split-unstable—even the real signal loses to a free one.

2 Related Work

Semantic entropy as a target. Our prediction target descends from work that measures uncertainty over meanings rather than tokens. Kuhn et al. (2023) introduced semantic entropy, clustering sampled generations into meaning-equivalence classes and computing entropy over those clusters, arguing that surface-level token probabilities conflate paraphrase with genuine disagreement. Farquhar et al. (2024) scaled this into a hallucination detector validated across tasks, establishing semantic entropy as a credible signal for content-level uncertainty. Because it requires multiple sampled generations and a clustering step, subsequent work seeks the same quantity more cheaply: Kossen et al. (2024) train probes on hidden states to predict semantic entropy in a single pass, while similarity-based methods refine or generalize the hard meaning-clustering step using continuous pairwise similarity (Nguyen et al., 2025; Nikitin et al., 2024). We adopt a hybrid equivalence construction (embedding cosine together with an NLI non-contradiction test) and use the resulting semantic entropy as our target. Our question sits one step beyond the trained probe: whether untrained geometric features recover this target at all, and whether they beat a trivial token-level signal in doing so. **Training-free geometric uncertainty.** A second line reads uncertainty from the geometry of hidden states, with no fitted probe. Trajectory-based methods treat the layer-wise path of a representation as informative: King et al. (2026) relate the curvature of this trajectory to next-token entropy, and Damirchi et al. (2026) use layer-wise geometric displacement to separate valid from spurious reasoning. Spectral methods instead summarize the hidden-state distribution—eigenscore (Chen et al., 2024) and effective rank (Wang et al., 2025) quantify how many directions the representation spreads across, on the intuition that a more diffuse spectrum signals uncertainty—while structural-confidence approaches fold spectral, local, and global trajectory descriptors into a single-pass score (Yang et al., 2026). These features are attractive because they are free at inference and were proposed, in part, to capture semantic structure that token-level confidence misses. They are the objects of our audit: we take their proposed signals at face value and ask whether they survive comparison with a trivial baseline and across evaluation protocols. A related family scores hallucinations from a single response using its internal attention-map and hidden-state statistics together with output probabilities, emphasizing compute efficiency (Sriramanan et al., 2024). Most directly, Yang et al. (2026) report that such structural descriptors, fused by a gradient-boosted model, beat probabilistic baselines including token entropy—the opposite of our headline; the settings differ in three respects (supervised fusion versus a single untrained statistic, a correctness target versus semantic entropy, and no multi-seed split test) that we make precise in the Discussion. **Length and surface-feature confounds.** A recurrent lesson in NLP evaluation is that a metric apparently tracking a target of interest is in fact tracking a surface property correlated with it. Benchmark artifacts—lexical overlap, answer position, and most persistently sequence length—let models or metrics score well for reasons unrelated to the intended construct (Søgaard et al., 2021). Uncertainty estimation is especially exposed: Santilli et al. (2025) prove and demonstrate that when a UQ method and its correctness function share a length bias, the interaction systematically distorts evaluation, and length bias afflicts sequence-level uncertainty directly (Gupta et al., 2024). Length-controlling by regression has been used to debias related evaluations (Dubois et al., 2024), a precedent for our quadratic length control. Our spectral findings are an instance of this family: two prominent geometric features are largely explained by sequence length and do not survive nonlinear length control. We state this correlationally and note the limitation that length and an upstream factor such as question difficulty are not separable in our data. **Evaluation-protocol fragility.** Two concurrent efforts already probe the reliability of geometric uncertainty signals. Yeats et al. (2025) ask what geometric internal-state metrics actually capture and find many are substantially sensitive to task-domain shift—enough to swamp the detection margin across domains. Aiersilan (2026), in a setup close to ours (Llama-3 and Mistral in 4-bit quantization, per-layer states, TruthfulQA), reports that a mid-network band carries most of the linearly decodable signal while sampling-based detectors underperform. Both suggest conclusions about these features depend heavily on how they are measured. This echoes a broader methodological literature: Søgaard et al. (2021) show that even held-out splits give optimistic, unstable estimates; further critiques argue the train–test paradigm does not insulate conclusions from reliability problems and that point estimates should give way to confidence intervals and multi-seed reporting (Xue et al., 2023; Riezler and Hagmann, 2022); and this fragility is sharpest for small effects on modest samples, where a single random split can mislead (An et al., 2021). Most evaluations of training-free uncertainty features nonetheless report one protocol on one split. We show that for geometric features at realistic held-out sample sizes this is not enough: the set that appear to carry signal changes substantially across resamplings of the same data—so a single-split verdict about which geometric features "work" is largely an artifact of evaluation design. We demonstrate that instability concretely, on this specific class of features, rather than as a general caution. A complementary result sits on a different axis: Orgad et al. (2025) show that detectors trained on internal states fail to generalize across datasets, concluding that truthfulness encoding is multifaceted rather than universal. Our instability is orthogonal to that cross-dataset transfer failure—it is within-distribution resampling variance in which untrained geometric features clear a bootstrap confidence interval at a fixed held-out size—so the two cautions stack rather than overlap.

3 Methods

Models and data. We study six model×dataset cells: two models—`meta-llama/Meta-Llama-3-8B-Instruct` and `mistralai/Mistral-7B-Instruct-v0.2`—each evaluated on TruthfulQA, BioASQ, and SQuAD v2. Both models are loaded with an identical `BitsAndBytesConfig` (`load_in_4bit=True`, `bnb_4bit_quant_type=nf4`, `bnb_4bit_use_double_quant=True`, `bnb_4bit_compute_dtype=float16`); we state this explicitly because identical quantization means any cross-model difference reflects the model, not its compression. Each of the three datasets contains 817 questions, split 572/245 (train/test) at seed 42, with this split applied uniformly to all six cells. For each question we draw ten independent generations using the model's chat template with the question as the sole user turn (`do_sample=True`, temperature 1.0, top-p 0.9, `max_new_tokens=50`, input truncation at 512 tokens), produced by ten separate decoding calls rather than batched return sequences. **Prediction target: hybrid semantic entropy.** The quantity each feature must predict is semantic entropy over the ten generations, following the meaning-clustering formulation of Kuhn et al. (2023) and Farquhar et al. (2024). Two generations join the same meaning cluster when they satisfy a hybrid equivalence relation: embedding cosine similarity (`all-mpnet-base-v2`) $\geq$ 0.90 and `roberta-large-mnli` returning a non-contradiction label in both directions. Clusters form by connected components over this relation, and semantic entropy is the entropy of the resulting cluster distribution. We hybridize two equivalence signals (embedding and NLI) because strict bidirectional NLI entailment alone fails to recognize obvious paraphrases at any usable threshold. This target is treated as a fixed ground truth, not proposed as a contribution; its coarse discretization is a limitation we quantify in the Limitations section. **The baseline geometry must beat: token entropy.** Our reference signal is the model's mean per-token predictive entropy—the entropy of the next-token distribution averaged over the question's forward pass, one scalar per question requiring no hidden states and no sampling. It is computed from the model's logits in the same forward pass used to extract the geometric features, and is wholly separate from the answer-sampling pipeline that defines the target. It is therefore not an ingredient of semantic entropy: the target is a hybrid of two equivalence relations with no token-entropy term, so the comparison is non-circular and any case where token entropy matches or beats a geometric feature is a fair result, not a shared-input artifact. **Geometric features.** Hidden states are extracted from the bare question only (truncation at 512 tokens), so every geometric feature—and the token-entropy baseline—is a property of the question's forward trajectory rather than of the sampled answers; the hypothesis under test is whether question-trajectory geometry predicts the uncertainty of the answer distribution. We save the hidden states at nine layers—$\{0, 4, 8, 12, 16, 20, 24, 28, 31\}$, i.e. every fourth layer from 0 through 28 plus the final layer (31), as both models have 32 transformer layers indexed 0–31. Curvature is the mean turning angle along the per-layer trajectory of mean-pooled hidden states over the pre-registered 0–16 sub-window (the first five of these: layers 0, 4, 8, 12, 16), with the per-layer standardization fit on training rows only, so no test information enters the feature. We additionally compute two prominent spectral features whose length-dependence we audit, taking care to use their exact forms. Eigenscore is computed on the full per-token, per-layer stack—all nine saved layers' per-token representations, mean-centered—as the Shannon entropy of the normalized squared singular values: with $s^2 = S^2$ and $p_i = s_i^2 / \sum_j s_j^2$, eigenscore $= -\sum_i p_i \log p_i$. Effective rank is computed on the mean-pooled per-layer trajectory—the 9×4096 matrix of layer means, mean-centered—as the exponential of the entropy of the raw normalized singular values: with $\hat{s}_n = S_n / \sum_m S_m$, effective rank $= \exp(-\sum_n \hat{s}_n \log \hat{s}_n)$. The two differ deliberately: eigenscore uses squared singular values and returns entropy directly on the full token stack, whereas effective rank uses raw singular values and returns the exponentiated entropy on the mean-pooled layer trajectory. **Local intrinsic dimensionality.** We also computed local intrinsic dimensionality (LID), estimated per question by local TwoNN over the layer-16 last-token neighborhood, with the neighbor tree built on training rows only. It is genuinely length-independent (per-cell correlation with sequence length -0.18 to $+0.22$) and weak: its partial association with semantic entropy includes zero in five of six cells at both endpoints. On the sixth, SQuAD-Mistral, it is small and *negative* with an interval excluding zero at both endpoints (-0.122 $[-0.183, -0.056]$ at full-817; -0.191 $[-0.288, -0.081]$ held-out). We record this for completeness rather than as support for LID as an uncertainty signal—the sign runs opposite to the hypothesis, the effect is confined to one of six cells, and an earlier, larger LID estimate on the anchor cell proved to be substantially neighbor-tree leakage, halving to a zero-crossing interval once the tree was restricted to training rows. We do not analyze LID further. **Confound controls and estimation.** All feature–target associations are partial correlations, with bootstrap 95% confidence intervals; a feature is "CI-clean" in a cell when its interval excludes zero. To remove length artifacts we control for sequence length and its square throughout—the nonlinear term matters because linear-only length control leaks signal that the quadratic term removes. Two analyses use these controls differently. To test whether a geometric feature carries signal beyond the free baseline, we report its partial correlation with semantic entropy controlling sequence length, length^2, and token entropy jointly. To ask which of curvature and token entropy better predicts semantic entropy, we compare them on equal footing: each is partialled for sequence length and length^2 only—the identical control set for both features, with neither feature controlled for the other. **Evaluation protocol and pre-registration.** The pre-registered primary endpoint is the held-out test set ($n = 245$, 30% split, seed 42); the full-sample evaluation ($n = 817$) is reported only as a power supplement and is explicitly not an out-of-sample claim, since it includes the rows on which features were characterized. To test whether held-out conclusions are reproducible rather than split-specific, we repeat the held-out evaluation across five random splits (seeds 42, 7, 123, 2024, 99) and report how the set of CI-clean cells varies. The curvature window 0–16 is fixed by pre-registration; a post-hoc layer sweep is reported separately and labeled as such, so that no window is selected after seeing its result.

4 Results

Setup recap. We report partial correlations between each feature and hybrid semantic entropy across the six cells, with bootstrap 95% confidence intervals.[1] The pre-registered primary endpoint is the held-out test set ($n = 245$, seed 42); full-sample results ($n = 817$) are a power supplement. Five-seed resampling probes reproducibility. Numbers below are keyed to Table 1 (per-cell baseline dominance), Figure 1 (length-confound), Figure 2 (curvature feature-fate), Table 2, and Figure 3 (the paired endpoint grid). **A free baseline matches or beats every geometric feature (C-B).** On equal footing (controlling sequence length and length^2 only, mean over five seeds), the model's mean per-token entropy out-predicts representational-trajectory curvature in every cell (Table 1). Token entropy wins on all five seeds in five of six cells; only on SQuAD-Mistral does it win 3/5, and there both features sit near zero. The margin (token entropy minus curvature) ranges from $+0.116$ to $+0.191$ across the five non-degenerate cells and its per-seed sign is stable across all five seeds except on that weakest cell. The point is not that token entropy is a strong predictor of semantic entropy in absolute terms—it is modest—but that a single free scalar requiring no hidden states is not beaten by the geometric feature introduced to improve on token-level signals. Geometry fails to beat its own baseline. The head-to-head in Table 1 is

Table 1: Equal-footing comparison: partial correlation of curvature and the token-entropy baseline with hybrid semantic entropy, each controlling sequence length and length^2 only (identical controls), mean over five seeds. Token entropy wins on all five seeds in five of six cells; on SQuAD-Mistral both features are near zero. The gap (token entropy $-$ curvature) ranges $+0.116$ to $+0.191$ on the five non-degenerate cells.

Cell	Curvature	Token entropy	Token wins
TQA-Llama	$+0.034$	$+0.224$	5/5
TQA-Mistral	$+0.011$	$+0.149$	5/5
BioASQ-Llama	$+0.077$	$+0.268$	5/5
BioASQ-Mistral	$+0.077$	$+0.195$	5/5
SQuAD-Llama	$+0.145$	$+0.261$	5/5
SQuAD-Mistral	$+0.071$	$+0.108$	3/5

[1] Curvature's primary held-out estimate controls for sequence length, length^2, and token entropy ($|$seq, seq^2, token), matching the headline figure. The curvature-vs-baseline head-to-head instead controls both features for sequence length and length^2 only ($|$seq, seq^2), so that geometry and the token-entropy baseline are compared on identical footing—one cannot control a feature for token entropy and then compare it against token entropy. The small difference in SQuAD-Llama's held-out estimate between regimes ($+0.172$ vs $+0.178$) reflects only the additional token-entropy term.

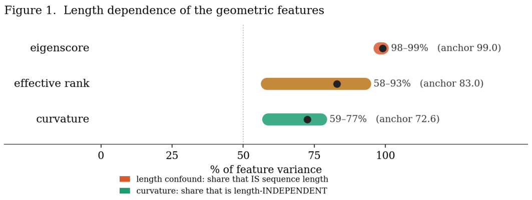

Figure 1: Length dependence of the geometric features. For eigenscore and effective rank, the bar shows the share of feature variance explained by sequence length and its square; for curvature, the complementary length-independent remainder. Eigenscore is almost entirely length (∼98–99%) and effective rank substantially so (58–93%), whereas the majority of curvature's variance (59–77%) is length-independent.

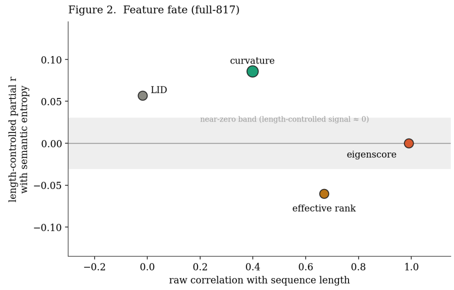

Figure 2: Feature fate (full-817): each geometric feature's raw correlation with sequence length (horizontal) against its length-controlled partial correlation with semantic entropy (vertical). Eigenscore and effective rank sit at high length-correlation and near-zero surviving signal; curvature alone retains a small length-independent association.

against curvature, the only geometric feature with a length-independent signal to compare; the prominent spectral features (eigenscore, effective rank) do not reach this comparison, because their association with semantic entropy is statistically null once sequence length and its square are controlled (see C-A below). Token entropy is thus not beaten by any feature we test. **Two prominent spectral features are sequence-length confounds (C-A).** This subsection explains why geometric features can appear useful before controls. Eigenscore is almost perfectly collinear with sequence length: its per-cell correlation with length runs from $+0.981$ to $+0.996$ ($\sim$98–99% of variance), and this length-coupling is independent of the semantic-entropy target. Once length is controlled, eigenscore's residual association with semantic entropy collapses toward zero, its confidence interval crossing zero. Effective rank is less extreme but tells the same story: sequence length and its square explain 58–93% of its variance (83% on the anchor cell), and its residual semantic-entropy association collapses toward zero under the same control. Critically, effective rank's apparent survival under linear-only length control disappears once the quadratic term is added—evidence that the nonlinear control is doing necessary work and that linear-only adjustment leaves a spurious length residual (Figure 1). We state this correlationally: sequence length is highly collinear with these features, but our data cannot separate length from an upstream factor such as question difficulty, so we do not claim length is the causal source. **The one length-independent feature is small, redundant, and split-unstable (C-D).** Curvature is the only geometric feature that survives the length audit—the majority of its variance is length-independent (non-length remainder 59–77% across cells; 72.6% on the anchor) and its small association persists under both linear and quadratic length control. But three facts bound how much this is worth. First, the effect is small: at full power the anchor cell (TQA-Llama) gives partial r $+0.086$ $[+0.018, +0.156]$. Second, at the pre-registered held-out endpoint it is CI-clean in only one of six cells—SQuAD-Llama, $+0.172$ $[+0.050, +0.297]$—while the other five include zero (Table 2, held-out column). We therefore describe SQuAD-Llama as CI-clean on the pre-registered split, never as robust. Third, as shown above, even this surviving signal is beaten by the free token-entropy baseline on five of five seeds in five of six cells. The one real geometric effect is genuine but redundant (Figure 2). **Which features "survive" is determined by evaluation design, not model family (C-C).** The audit's central instability appears when the same six cells are scored at two endpoints (Table 2, Figure 3). At full statistical power ($n = 817$), four cells are CI-clean: TQA-Llama ($+0.086$ $[+0.018, +0.156]$), BioASQ-Llama ($+0.087$ $[+0.018, +0.153]$), BioASQ-Mistral ($+0.090$ $[+0.021, +0.160]$), and SQuAD-Llama ($+0.100$ $[+0.029, +0.168]$). At the pre-registered held-out endpoint ($n = 245$), the number of CI-clean cells is not four but varies with the random split: across seeds 42, 7, 123, 2024, 99 the clean count is 1, 0, 3, 2, 0 respectively (mean 1.2, range 0–3), and no cell is clean on all five splits. On the displayed split (seed 42) only SQuAD-Llama survives; on two seeds nothing survives; on one seed three cells survive, including cells the seed-42 figure shows as null. The survivor set is therefore largely a resampling artifact at this sample size, and a single held-out verdict about which geometric features work is not reproducible. The BioASQ cells make the fork explicit, and we present it transparently rather than choosing the flattering reading. At full power BioASQ-Mistral is CI-clean and positive ($+0.090$ $[0.021, 0.160]$); at the held-out endpoint the same cell is null and slightly negative (-0.011 $[-0.140, 0.122]$). Two readings are available. Under the first, the full-sample positives are real and the held-out nulls are merely underpowered, so curvature is a small but fairly general effect sitting near the $n = 245$ detection floor. Under the second, the full-sample column is optimistic because it includes the rows on which features were characterized, and the held-out column is the trustworthy out-of-sample estimate, which is null off the anchor. We do not adjudicate between them. Because the held-out endpoint is the pre-registered primary, we default to the conservative reading—curvature's effect is not reliably recoverable out-of-sample beyond the anchor—while reporting the full-817 column in full so the reader can weigh the alternative. We emphasize that BioASQ-Mistral's full-to-held-out reversal is an instance of endpoint instability—a cell that is clean at full power going null out-of-sample—and not a model-family effect; the same instability afflicts cells in both models, which is precisely the point.

5 Discussion

What the audit shows. Taken together, the three findings describe a training-free geometric uncertainty signal that does not hold up the way its motivation suggests. A free token-level scalar is not beaten by any geometric feature we test (C-B); the two spectral features that appear useful—eigenscore and effective rank—are largely sequence length and lose their semantic-entropy association once length is controlled nonlinearly (C-A); the single feature with a genuine length-independent signal, curvature, is small and redundant with that free baseline (C-D); and the set of features that appear to "survive" at a held-out endpoint is unstable across resampling, ranging from zero to three CI-clean cells where the full-sample evaluation is fixed at four (C-C). The unifying lesson is methodological rather than mechanistic: conclusions about training-free geometric uncertainty features are fragile to evaluation design, and a single-protocol, single-split report can read as a positive result that a reseeded held-out evaluation does not reproduce. **What this does, and does not, claim about geometry.** Our result is deliberately relative. Token entropy is itself only a modest predictor of semantic entropy in absolute terms; we do not claim it is a strong detector, nor that hidden states contain no uncertainty information. The claim is narrower and, we think, more useful: the geometric features were introduced to improve on token-level signals, and as untrained direct predictors of semantic entropy they do not. This is consistent with—and does not contradict—work showing that probes trained on hidden states recover semantic entropy (Kossen et al., 2024); learning a mapping from representations is a different and more permissive setting than reading a fixed geometric statistic. Our negative is specific to the training-free, geometric-feature-as-predictor case. **Relation to concurrent audits.** Two recent studies reach compatible conclusions from different angles. Yeats et al. (2025) find that geometric internal-state metrics are sensitive to task-domain shift; Aiersilan (2026), in a quantized setup close to ours, finds that the linearly decodable hallucination signal concentrates in a mid-network band while sampling-based detectors underperform. Our contribution is complementary: an equal-footing comparison against a trivial baseline, and a direct quantification of how unstable the held-out verdict is across random splits at a realistic sample size. The three results converge on the same caution rather than competing. **Reconciling an apparent counterexample.** Yang et al. (2026) report that single-pass structural descriptors beat probabilistic baselines, token entropy among them, which appears to contradict our headline. It does not, for three reasons. First, their score is supervised: a gradient-boosted model is trained to fuse many spectral, local, and global descriptors, whereas we read a single untrained geometric statistic—learning a fused mapping is the more permissive setting we already distinguish from reading a fixed statistic. Second, their prediction target is answer correctness, while ours is semantic entropy; a feature can track correctness without tracking meaning-level uncertainty. Third, their evaluation does not probe multi-seed split stability, the very axis on which we find the held-out verdict fragile. A trained fusion that beats token entropy at predicting correctness is therefore compatible with an untrained single statistic that does not beat token entropy at predicting semantic entropy on equal footing. **Recommendation.** The constructive reading is straightforward. Anyone

Table 2: Curvature's partial correlation with semantic entropy at two endpoints, controlling sequence length, length^2, and token entropy. Full-817 is the powered (in-sample) supplement; held-out ($n = 245$, seed 42) is the pre-registered primary. Four cells are CI-clean at full power; on the displayed held-out split only SQuAD-Llama is. The two cells not CI-clean at full power are shown as point estimates. "Clean?" refers to the held-out endpoint on this split; across five seeds the held-out clean count is 0–3 (see text).

Cell	Full-817 r [95% CI]	Held-out r [95% CI]	Clean?
TQA-Llama	$+0.086$ $[0.018, 0.156]$	$+0.098$ $[-0.041, 0.228]$	no
SQuAD-Llama	$+0.100$ $[0.029, 0.168]$	$+0.172$ $[0.050, 0.297]$	yes
BioASQ-Llama	$+0.087$ $[0.018, 0.153]$	$+0.053$ $[-0.076, 0.180]$	no
BioASQ-Mistral	$+0.090$ $[0.021, 0.160]$	-0.011 $[-0.140, 0.122]$	no
TQA-Mistral	$+0.055$	$+0.044$ $[-0.085, 0.174]$	no
SQuAD-Mistral	$+0.039$	$+0.030$ $[-0.086, 0.145]$	no

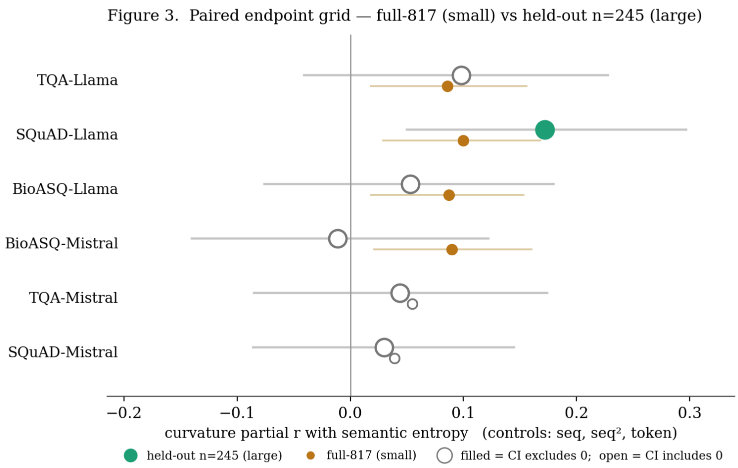

Figure 3: Curvature's residual partial correlation with semantic entropy across the six model×dataset cells, paired at two endpoints (small marker: full-817; large marker: held-out $n = 245$, seed 42; controls: sequence length, length^2, token entropy; filled: 95% CI excludes zero, open: includes zero). Four cells are CI-clean at full power; the held-out CI-clean count ranges 0–3 across seeds 42, 7, 123, 2024, 99, with none clean on all five—so the surviving set is largely a function of evaluation design, not model family (cf. Table 2).

proposing or reviewing a training-free geometric uncertainty feature should (i) compare it against a token-entropy baseline on equal footing—identical controls for both—since a feature introduced to beat token-level signals should be shown to do so; (ii) report multi-seed held-out confidence intervals rather than a single-split point estimate, because the survivor set at these sample sizes is partly a resampling artifact; and (iii) control sequence length nonlinearly, as linear-only adjustment leaves a spurious residual that can masquerade as signal. Applied to the features here, these three checks are what turn an apparent positive into the negative we report. The broader point is that the cost of each check is trivial relative to the cost of a misattributed geometric "signal"; a free baseline and a handful of extra seeds are enough to separate a real effect from an artifact of evaluation design.

Limitations

Our scope is bounded in ways a different study might probe further: two models, three datasets, one curvature definition (mean turning angle over the 0–16 window), one hybrid semantic-entropy construction, and one token-entropy baseline. A different geometric feature or a different uncertainty target could behave differently—though we note that the instability finding (C-C) concerns evaluation design and is largely independent of which specific feature is measured. Our baseline is question-position predictive entropy, computed from the question's forward pass; a generation-derived token entropy is the more conventional choice and might be stronger still, which would only deepen the negative, but we have not tested it. And, as stated in the results, sequence length is highly collinear with the spectral features but is not separable in our data from an upstream factor such as question difficulty, so our length-confound claim is correlational, not causal. Two further limitations are methodologically load-bearing and we state both directions honestly. First, at full statistical power several cells—including both BioASQ cells—show small CI-clean positive curvature effects, which the held-out endpoint does not reproduce; we default to the conservative held-out reading on pre-registration grounds, but the small-but-general reading is available and we do not foreclose it. Relatedly, the held-out sample size ($n = 245$) sits near the detection floor for effects this small: this is simultaneously part of the C-C instability finding and a genuine limit on our power to affirm a weak geometric signal, and we do not want either edge of that to be read alone. Second, the mid-network layer band where curvature peaks is a post-hoc, cell-specific observation; we report the pre-registered 0–16 window as primary precisely to avoid selecting a window after seeing its result, and a pre-registered mid-band study would be the right way to revisit it. A third limitation concerns the target itself, and it bounds how every magnitude in this paper should be read. Hybrid semantic entropy over ten generations can take only the finitely many values attainable by partitions of ten items, and our data occupies that grid very unevenly. Each cell realizes 34–37 distinct values (38 pooled across the six cells), and the ceiling $\ln 10 \approx 2.303$—all ten generations in separate meaning clusters—is reached on only 1.2–6.7% of items. The dominant feature of the distribution is instead a large point mass at exactly zero, where all ten generations merge into a single cluster: 21.9% of items on SQuAD-Llama, 50.1% on BioASQ-Mistral, and 36.0% pooled across the grid. Partial correlations computed against a target carrying more than a third of its mass at a single floor value are attenuated, so the absolute size of every association we report—geometric and baseline alike—is a lower bound rather than a calibrated effect size. This cuts two ways for our conclusions. It does not threaten the equal-footing comparison (C-B): curvature and token entropy face the identical target, so the ordering between them is unaffected by the floor mass. It does, however, plausibly contribute to the held-out instability we document (C-C), since a heavily floored target at $n = 245$ is precisely the regime in which CI-cleanness should be expected to flicker across resamplings—and we cannot separate that contribution from resampling variance itself. A finer-grained target, whether through more samples per question or a continuous similarity-based construction (Nguyen et al., 2025; Nikitin et al., 2024), is the direct way to reduce this floor mass, and we regard it as the most informative single extension of this work. Even the geometric feature with a real, length-independent signal is beaten by a free baseline and is not reliably recoverable out-of-sample beyond a single cell. The practical conclusion is not that hidden-state geometry is uninformative, but that demonstrating a training-free geometric uncertainty feature requires clearing a trivial baseline and surviving a reseeded held-out evaluation—two bars that are cheap to test and that the features examined here do not clear. **Reproducibility and availability.** The estimates, confidence intervals, and seed-level counts reported above are drawn from result files fixed at analysis time. Those files, the feature-extraction and statistical analysis code, and the per-question feature table—semantic entropy, mean per-token entropy, sequence length, and each geometric feature—will be released under an open license. Pending that release, the grid is reconstructible from the specification in Section 3: both models are public checkpoints loaded under an identical 4-bit NF4 configuration with double quantization and `float16` compute dtype; the train/test split is 572/245 at seed 42; the resampling seeds are 42, 7, 123, 2024 and 99; the curvature window is layers 0–16, fixed by pre-registration; and the semantic-entropy target is the hybrid construction of embedding cosine ≥ 0.90 with a bidirectional NLI non-contradiction test.

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